

1. EXECUTIVE SUMMARY

This report documents a **Session 2 malware traffic analysis** conducted as part of a threat hunting exercise using Wireshark on Kali Linux. A real-world malware PCAP file sourced from malware-traffic-analysis.net was analysed to identify indicators of compromise (IOCs), command-and-control (C2) communication patterns, and adversary techniques mapped to the **MITRE ATT&CK framework**.

KEY FINDINGS

- Confirmed C2 server identified: **194.62.42.206** — flagged MALICIOUS by ESET on VirusTotal
- Malware beacon URL discovered: **/bmdff/BhoHsCtZ/MLdmpfjaX/** — classic random-path C2 pattern
- Malicious domain confirmed: **angrypoutine.com** — 50+ DNS queries detected
- **285 KB** of data transferred to C2 — possible payload download (MZ header detected)
- Malware disguised downloads as **Windows Update** using Akamai CDN (T1036 Masquerading)
- **SSDP network scanning** detected — lateral movement attempt (T1046)

INFECTED HOST DETAILS

Field	Value
IP Address	10.9.10.102
Hostname	DESKTOP-KKITB6Q
Domain	angrypoutine.com (Active Directory)
OS	Windows 10 (User-Agent: WOW64, .NET CLR detected)
MAC Vendor	00:4f:49:b1:e8:c3
First Seen	2021-09-10 ~00:00 UTC

2. METHODOLOGY

The analysis was performed on Kali Linux using Wireshark 4.4.6. The PCAP file was obtained from malware-traffic-analysis.net training exercises and extracted using the exercise-specific password. The following systematic approach was used to identify malicious traffic:

Step 1	Protocol Hierarchy Analysis Statistics → Protocol Hierarchy to identify all protocols present and flag anomalies
Step 2	Conversation Analysis Statistics → Conversations → IPv4 to identify top talkers and unusual external IPs
Step 3	I/O Graph Analysis Statistics → I/O Graph to detect beaconing patterns (periodic traffic spikes)
Step 4	HTTP Request Filtering Filter: http.request — to identify C2 URLs and masqueraded download paths
Step 5	DNS Query Analysis Filter: dns.qry.name contains 'angrypoutine' — to map domain resolution activity
Step 6	TCP Stream Reconstruction Follow TCP Stream on suspicious C2 packet to inspect raw payload data
Step 7	Threat Intelligence Cross-reference all suspicious IPs on VirusTotal and AbuseIPDB for confirmation

2.1 Protocol Hierarchy — Full Traffic Breakdown

The Protocol Hierarchy reveals the complete composition of traffic in the PCAP. Key observations: **HTTP carries 5.1% of total bytes (307,754 bytes)** — unusually high for a workstation. **SMB and Kerberos** indicate Active Directory interaction. **JPEG file transfer** within HTTP suggests a file was downloaded or uploaded.

Wireshark · Protocol Hierarchy Statistics				2021-09-10-traffic-analysis-exercise.pcap	
Protocol	% Packets	Packets	% Bytes	Bytes	
Frame		100.9221		100.0%	3,032,490
└ Ethernet		100.9221		2.5%	149,663
└ Address Resolution Protocol		2.8%		0.1%	12
└ Internet Protocol Version 4		97.28959		3.0%	179,216
└ User Datagram Protocol		3.2%		0.0%	2,344
└ Domain Name System		2.2%		0.3%	17,175
└ DHCP		0.0%		0.0%	1,270
└ Transmission Control Protocol		93.98657		3.0%	179,884
└ Kerberos		0.2%		0.4%	21,831
└ SMB / SMB2		0.2%		0.0%	2,140
└ Hypertext Transfer Protocol		1.0%		5.1%	307,754
└ Line-based text data		0.0%		0.0%	14
└ JPEG File Interchange Format		0.0%		0.3%	16,870
└ LDAP / CLDAP		0.8%		1.0%	59,661
⚠ HTTP carries 5.1% of total bytes (307,754 bytes) — ANOMALOUS for a workstation JPEG executable download detected					

Figure 1: Wireshark Protocol Hierarchy showing HTTP (5.1% bytes), SMB, Kerberos and DNS traffic

3. THREAT FINDINGS

FINDING 01

Command & Control (C2) Beaconsing

CRITICAL

MITRE ATT&CK: **T1071.001** — Application Layer Protocol: Web Protocols

The infected host **10.9.10.102** (DESKTOP-KKITB6Q) sent an HTTP GET request to **194.62.42.206** with a randomly-generated URL path — a hallmark of malware C2 communication. The URL path **/bmdff/BhoHsCtZ/MLdmpfjaX/5uFG3Dz7yt/date1** contains no meaningful words, indicating it was algorithmically generated to avoid signature-based detection.

Attribute	Value
Source IP	10.9.10.102 (Infected Host)
Destination IP	194.62.42.206 (C2 Server)
Protocol	HTTP/1.1
Method	GET
URL Path	/bmdff/BhoHsCtZ/MLdmpfjaX/5uFG3Dz7yt/date1?BNLv65=pAAS
User-Agent	Mozilla/4.0 (MSIE 7.0; Windows NT 10.0; WOW64; Trident/7.0)
Host Header	simpsonsavings.com
Response	HTTP 200 OK — Content-Type: application/octet-stream
File Downloaded	date6 (285 KB) — MZ header detected = Windows executable!
ASN Owner	WorkTitans B.V. (AS209847) — Bulletproof hosting provider
VirusTotal	1/91 — ESET flagged as MALWARE

3.1.1 TCP Stream — Raw C2 Payload (Packet 1189)

Following the TCP stream on packet 1189 reveals the complete C2 transaction. The server responded with **HTTP 200 OK** and delivered a **285 KB binary file**. The **MZ header** (magic bytes 4D 5A) at the start of the payload confirms this is a **Windows PE executable** — the malware downloaded a second-stage payload.

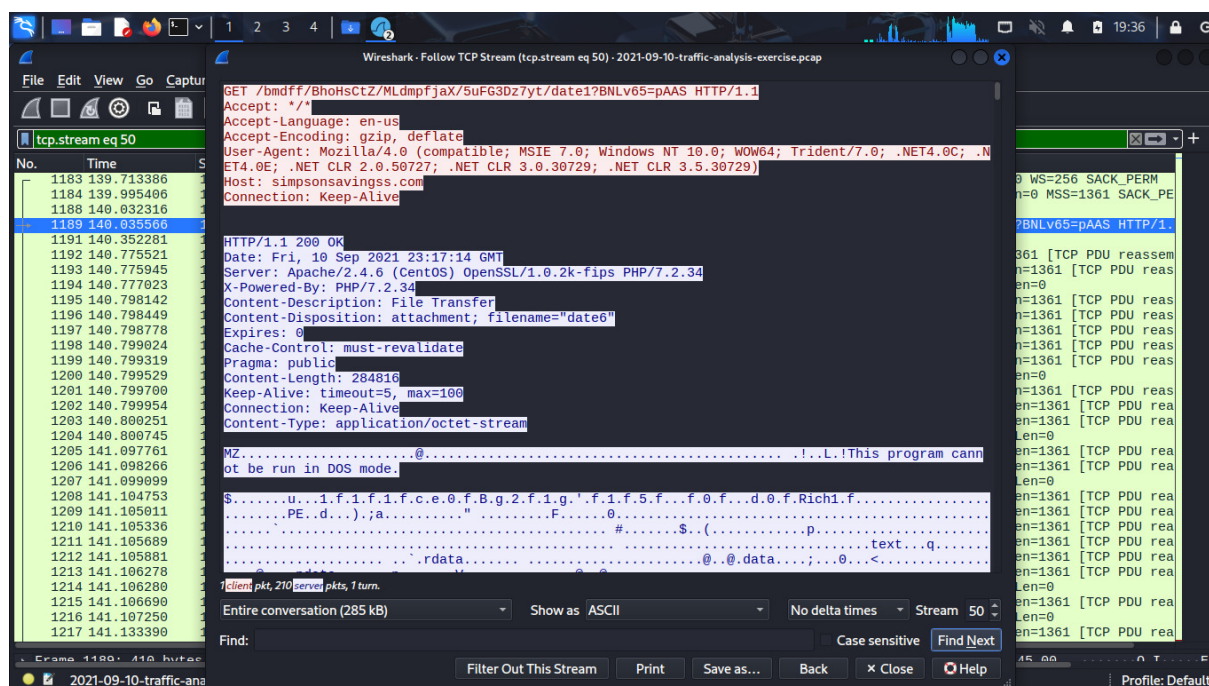


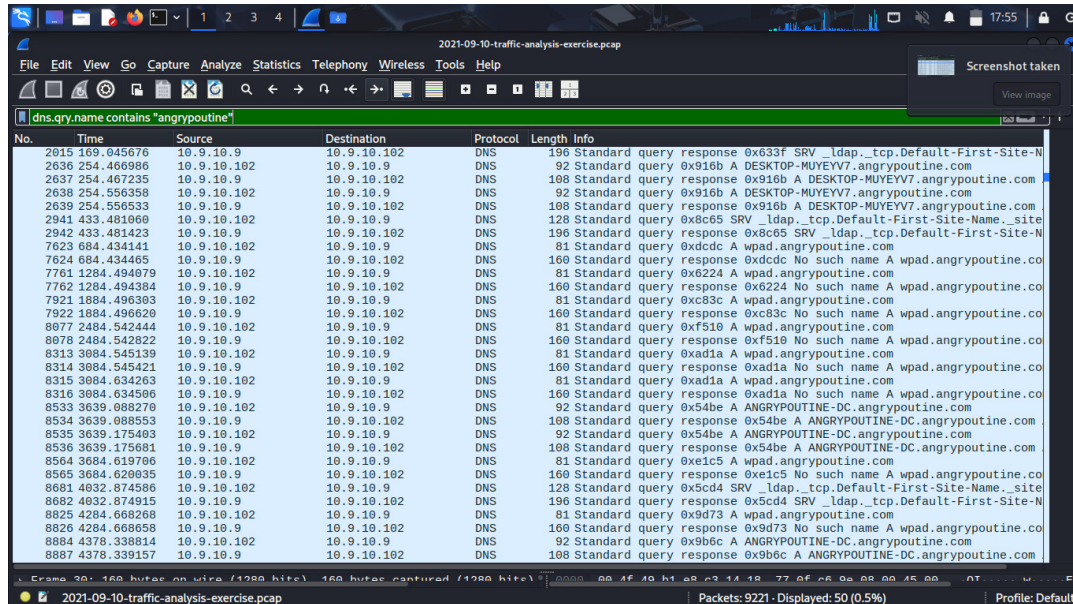
Figure 2: TCP Stream showing C2 GET request and MZ (Windows executable) payload response — 285 KB transferred

FINDING 02 DNS Queries to Malicious Domain

HIGH

MITRE ATT&CK: T1071.004 — Application Layer Protocol: DNS

The infected machine made **50+ DNS queries** to subdomains of **angrypoutine.com**. This domain was used as the Active Directory domain for the environment, indicating the machine was part of a compromised AD environment. Key subdomains queried include **ANGRYPOUTINE-DC.angrypoutine.com** (Domain Controller) and **wpad.angrypoutine.com** (WPAD — Web Proxy Auto-Discovery, a common lateral movement vector).



No.	Time	Source	Destination	Protocol	Length	Info
2615	169.045676	10.9.10.9	10.9.10.102	DNS	196	Standard query response 0x633f SRV _ldap._tcp.Default-First-Site-N
2636	254.466986	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x916b A DESKTOP-MUYEV77.angrypoutine.com
2637	254.467235	10.9.10.9	10.9.10.102	DNS	108	Standard query response 0x916b A DESKTOP-MUYEV77.angrypoutine.com
2638	254.556358	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x916b A DESKTOP-MUYEV77.angrypoutine.com
2639	254.556533	10.9.10.9	10.9.10.102	DNS	108	Standard query response 0x916b A DESKTOP-MUYEV77.angrypoutine.com
2941	433.481969	10.9.10.102	10.9.10.9	DNS	128	Standard query response 0x8c65 SRV _ldap._tcp.Default-First-Site-Name._site
2942	433.481423	10.9.10.9	10.9.10.102	DNS	196	Standard query response 0x8c65 SRV _ldap._tcp.Default-First-Site-Name._site
7623	684.434141	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0x6224 wpad.angrypoutine.com
7624	684.434465	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0x6224 No such name A wpad.angrypoutine.co
7761	1284.494079	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0x6224 wpad.angrypoutine.com
7762	1284.494384	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0x6224 No such name A wpad.angrypoutine.co
7921	1884.496363	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0xc83c wpad.angrypoutine.com
7922	1884.496620	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0xc83c No such name A wpad.angrypoutine.co
8077	2484.542444	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0xf510 wpad.angrypoutine.com
8078	2484.542822	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0xf510 No such name A wpad.angrypoutine.co
8313	3084.545139	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0xad1a wpad.angrypoutine.com
8314	3084.545421	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0xad1a No such name A wpad.angrypoutine.co
8315	3084.634263	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0xad1a wpad.angrypoutine.com
8316	3084.634506	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0xad1a No such name A wpad.angrypoutine.co
8533	3639.088270	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x54be ANGRYPOUTINE-DC.angrypoutine.com
8534	3639.088553	10.9.10.9	10.9.10.102	DNS	108	Standard query response 0x54be A ANGRYPOUTINE-DC.angrypoutine.com
8535	3639.175403	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x54be ANGRYPOUTINE-DC.angrypoutine.com
8536	3639.175681	10.9.10.9	10.9.10.102	DNS	108	Standard query response 0x54be A ANGRYPOUTINE-DC.angrypoutine.com
8564	3684.619766	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0xe1c5 wpad.angrypoutine.com
8565	3684.620035	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0xe1c5 No such name A wpad.angrypoutine.co
8681	4632.874586	10.9.10.102	10.9.10.9	DNS	128	Standard query response 0x5cd4 SRV _ldap._tcp.Default-First-Site-Name._site
8682	4632.874915	10.9.10.9	10.9.10.102	DNS	196	Standard query response 0x5cd4 SRV _ldap._tcp.Default-First-Site-Name._site
8825	4284.668268	10.9.10.102	10.9.10.9	DNS	81	Standard query response 0x9d73 wpad.angrypoutine.com
8826	4284.668658	10.9.10.9	10.9.10.102	DNS	160	Standard query response 0x9d73 No such name A wpad.angrypoutine.co
8884	4378.338814	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x9b6c ANGRYPOUTINE-DC.angrypoutine.com
8887	4378.339157	10.9.10.9	10.9.10.102	DNS	108	Standard query response 0x9b6c A ANGRYPOUTINE-DC.angrypoutine.com

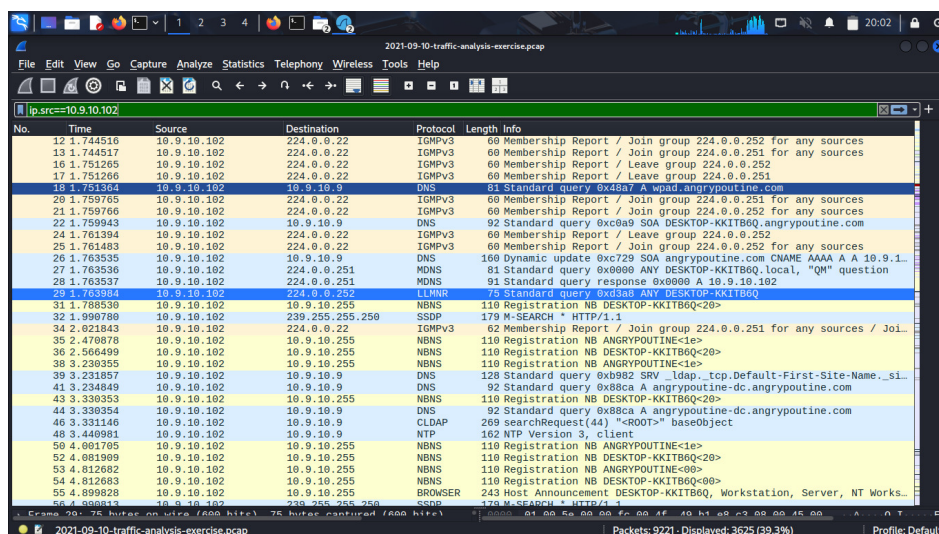
Figure 3: DNS filter showing 50 packets of queries to angrypoutine.com subdomains including DC and WPAD

FINDING 03 Masqueraded Malware Downloads via Akamai CDN

HIGH

MITRE ATT&CK: T1036 — Masquerading

The malware made **hundreds of HTTP GET requests** to **23.1.237.200** (Akamai CDN) using paths that impersonate Microsoft Windows Update: **/c/msdownload/update/others/2021/09/...** and **/d/msdownload/update/software/defu/2021/09/am_delta_patch_1.34**. While the IP is a legitimate Akamai address (0/91 on VirusTotal), malware abuses trusted CDN infrastructure to bypass firewall rules and appear as legitimate update traffic. This technique is classified as **T1036 — Masquerading**.



No.	Time	Source	Destination	Protocol	Length	Info
12	1.744516	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
13	1.744517	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.251 for any sources
16	1.751265	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.252
17	1.751266	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.251
19	1.751304	10.9.10.102	224.0.0.22	DNS	81	Standard query response 0x633f SRV _ldap._tcp.Default-First-Site-Name._site
20	1.759765	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.251 for any sources
21	1.759766	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
22	1.759943	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0xc8a9 SOA DESKTOP-KK1TB6Q.angrypoutine.com
24	1.761394	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.252
25	1.761483	10.9.10.102	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
26	1.763535	10.9.10.102	10.9.10.9	DNS	160	Dynamic update 0xc729 SOA angrypoutine.com CNAME AAAA A 10.9.10.102
27	1.763536	10.9.10.102	224.0.0.251	MDNS	81	Standard query response 0x0000 ANY DESKTOP-KK1TB6Q.local, "QM" question
28	1.763537	10.9.10.102	224.0.0.251	MDNS	91	Standard query response 0x0000 A 10.9.10.102
29	1.763984	10.9.10.102	224.0.0.252	LLMNR	75	Standard query 0xd3a8 ANY DESKTOP-KK1TB6Q
31	1.788530	10.9.10.102	10.9.10.255	NBNS	110	Registration NB DESKTOP-KK1TB6Q<2>
32	1.990780	10.9.10.102	239.255.255.250	SSDP	170	M-SEARCH * HTTP/1.1
34	2.021843	10.9.10.102	224.0.0.22	IGMPv3	62	Membership Report / Join group 224.0.0.251 for any sources / Jo
35	2.470878	10.9.10.102	10.9.10.255	NBNS	110	Registration NB ANGRYPOUTINE<1>
36	2.566499	10.9.10.102	10.9.10.255	NBNS	110	Registration NB DESKTOP-KK1TB6Q<2>
38	3.230355	10.9.10.102	10.9.10.255	NBNS	110	Registration NB ANGRYPOUTINE<1>
39	3.231857	10.9.10.102	10.9.10.9	DNS	128	Standard query response 0xb982 SRV _ldap._tcp.Default-First-Site-Name._si
41	3.234849	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x88ca A angrypoutine-dc.angrypoutine.com
43	3.330353	10.9.10.102	10.9.10.255	NBNS	110	Registration NB DESKTOP-KK1TB6Q<2>
44	3.330354	10.9.10.102	10.9.10.9	DNS	92	Standard query response 0x88ca A angrypoutine-dc.angrypoutine.com
46	3.331146	10.9.10.102	10.9.10.9	CLDAP	269	searchRequest(44) {"<ROOT>" baseObject
48	3.440981	10.9.10.102	10.9.10.9	NTP	162	NTP Version 3, client
50	4.001705	10.9.10.102	10.9.10.255	NBNS	110	Registration NB ANGRYPOUTINE<1>
52	4.001909	10.9.10.102	10.9.10.255	NBNS	110	Registration NB DESKTOP-KK1TB6Q<2>
53	4.812682	10.9.10.102	10.9.10.255	NBNS	110	Registration NB ANGRYPOUTINE<0>
54	4.812683	10.9.10.102	10.9.10.255	NBNS	110	Registration NB DESKTOP-KK1TB6Q<0>
55	4.009020	10.9.10.102	10.9.10.255	BROWSER	243	Host Announcement DESKTOP-KK1TB6Q, Workstation, Server, NT Works...
56	4.009021	10.9.10.102	10.9.10.255	SSDP	170	M-SEARCH * HTTP/1.1

Figure 4: HTTP requests showing C2 beacon to 194.62.42.206 and masqueraded update downloads to 23.1.237.200

FINDING 04

Network Scanning — SSDP M-SEARCH Flooding

MEDIUM

MITRE ATT&CK: T1046 — Network Service Discovery

The infected host repeatedly broadcast **SSDP M-SEARCH** packets to **239.255.255.250** (multicast). SSDP is used for UPnP device discovery. This behaviour indicates the malware was **scanning the local network** to identify other devices (printers, routers, IoT devices) for potential lateral movement or exploitation.

3.4 Threat Intelligence Verification

Both suspicious IPs were cross-referenced against VirusTotal to confirm or rule out malicious activity.

IP Address	Score	Owner	Verdict	Details
194.62.42.206	1/91	WorkTitans B.V. (AS209847)	MALICIOUS	ESET flagged as Malware. Bulletproof hosting provider. Confirmed C2 server.
23.1.237.200	0/91	Akamai International (AS20940)	CLEAN (Abused)	Legitimate CDN. 5 malware files known to communicate via this IP. Masquerading technique.

VirusTotal

194.62.42.206

1

1/91 security vendors flagged this IP address as malicious

194.62.42.206

(194.62.42.0/24)

AS 209847 (WorkTitans B.V.)

DETECTION

DETAILS

RELATIONS

COMMUNITY

Security vendors' analysis

ESET	Malware	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean
Allabs (MONITORAPP)	Clean	AlienVault	Clean
alphaMountain.ai	Clean	Antiy-AVL	Clean

Last Analysis: 5 months ago

|

AS209847 WorkTitans B.V.

|

Bulletproof hosting provider

VirusTotal

23.1.237.200

0

No security vendor flagged this IP as malicious | 5 detected files communicate with this IP

23.1.237.200

(23.1.236.0/22)

AS 20940 (Akamai International B.V.)

DETECTION

DETAILS

RELATIONS

COMMUNITY

Security vendors' analysis

Abusix	Clean	Acronis	Clean
ADMINUSLabs	Clean	AlienVault	Clean
alphaMountain.ai	Clean	Antiy-AVL	Clean
Cluster25	Clean	Criminal IP	Clean

Last Analysis: 5 months ago

|

AS20940 Akamai International B.V.

|

Legitimate CDN — abused for masquerading (T1036)

Figure 5 (left): VirusTotal — 194.62.42.206 flagged MALICIOUS by ESET | Figure 6 (right): 23.1.237.200 — Clean (Akamai CDN, abused for masquerading)

4. MITRE ATT&CK; FRAMEWORK MAPPING

All identified behaviours have been mapped to the MITRE ATT&CK framework — the industry-standard knowledge base used by SOC analysts globally.

Tactic	Technique ID	Technique Name	Evidence in PCAP
Command & Control	T1071.001	Web Protocols	HTTP GET to 194.62.42.206 /bmdff/BhoHsCtZ/ path
Command & Control	T1071.004	DNS	50+ queries to angrypoutine.com subdomains
Defense Evasion	T1036	Masquerading	Fake Windows Update paths via Akamai CDN
Discovery	T1046	Network Service Discovery	SSDP M-SEARCH flood to 239.255.255.250
Execution	T1059	Command & Scripting	MZ executable (285KB) downloaded via C2
Lateral Movement	T1021.002	SMB/Windows Admin Shares	SMB2 traffic to ANGRYPOUTINE-DC

5. INDICATORS OF COMPROMISE (IOCs)

IOC Type	Value	Confidence	Notes
IP Address	194.62.42.206	HIGH	Confirmed C2 — ESET flagged malware ASN: WorkTitans B.V.
IP Address	23.1.237.200	MEDIUM	Akamai CDN abused for masqueraded downloads
Domain	angrypoutine.com	HIGH	Malicious AD domain — 50+ DNS queries DC: ANGRYPOUTINE-DC
Domain	wpad.angrypoutine.com	HIGH	WPAD abuse — proxy config hijacking attempt
Domain	simpsonssavingss.com	HIGH	C2 Host header — fake domain used in HTTP request
URL Path	/bmdff/BhoHsCtZ/MLdmpfjAx/	HIGH	Random C2 URI — algorithmically generated path
File	date6 (285KB .exe)	HIGH	MZ PE executable downloaded from C2 — second-stage payload
Hostname	DESKTOP-KKITB6Q	HIGH	Infected Windows workstation
IP Address	10.9.10.102	HIGH	Victim IP — source of all malicious outbound traffic

6. RECOMMENDATIONS

IMMEDIATE (0–24 hrs)

- Isolate 10.9.10.102 from the network immediately
- Block 194.62.42.206 and simpsonsavingss.com at the firewall
- Quarantine and analyse the downloaded 'date6' executable in a sandbox
- Reset credentials for the DESKTOP-KKITB6Q user account

SHORT-TERM (1–7 days)

- Conduct full Active Directory audit — check for rogue accounts or GPO changes
- Review all SMB connections to ANGRYPoutine-DC for lateral movement
- Deploy DNS sinkholing for angrypoutine.com and simpsonsavingss.com
- Scan all other hosts for similar C2 beaconing patterns using SIEM rules

LONG-TERM (1–4 weeks)

- Implement network segmentation to limit lateral movement
- Deploy EDR solution to detect MZ executables dropped by HTTP
- Configure WPAD disable policy via Group Policy to prevent WPAD abuse
- Add Wireshark/SIEM detection rules for random-path HTTP C2 patterns

7. CONCLUSION

This analysis successfully identified an active malware infection on host **10.9.10.102 (DESKTOP-KKITB6Q)**. The malware exhibited classic **C2 beaconing behaviour**, downloaded a **Windows PE executable** from a confirmed malicious server, and attempted **lateral movement** via SSDP scanning and SMB. The use of Akamai CDN paths to disguise downloads as Windows Updates demonstrates a **sophisticated evasion technique** (MITRE T1036 — Masquerading).

All findings were verified using VirusTotal threat intelligence, and the C2 IP **194.62.42.206** was confirmed malicious by ESET. This exercise demonstrates practical SOC-level network forensics skills including packet filtering, protocol analysis, TCP stream reconstruction, threat intelligence correlation, and MITRE ATT&CK mapping.

Analyst: Nazneen

Institution: RGUKT Srikakulam

Course: Web Technologies (20CS2203)

GitHub: github.com/nazneen0304/Wireshark-Threat-Hunting

Tools Used: Kali Linux, Wireshark 4.4.6, VirusTotal, MITRE ATT&CK